

Module Title: STP Concepts

Module Objective: Explain how STP enables redundancy in a Layer 2 network.

Topic Title	Topic Objective
Purpose of STP	Explain common problems in a redundant, L2 switched network.
STP Operations	Explain how STP operates in a simple switched network.
Evolution of STP	Explain how Rapid PVST+ operates.

Page #	Activity Type	Activity Name
5.1.8	Video	Observe STP Operation
5.1.9	Packet Tracer	Investigate STP Loop Prevention
5.1.10	Check Your Understanding	Purpose of STP
5.2.12	Check Your Understanding	STP Operations

5.3.6

Check Your
Understanding

Evolution of STP

New Terms

Spanning Tree Protocol (STP)

Spanning Tree Algorithm (STA)

IEEE 802.1D

IEEE 802.1w

Broadcast Storm

Root Bridge

Root Port

Designated Port

Alternate (Blocked) Port

Learning

Listening

Bridge ID (BID)

Root ID

Bridge Protocol Data Unit (BPDU)

Bridge Priority

Extended System ID

short path cost

long path cost

root path cost

Rapid STP (RSTP)

port priority

Hello timer

Max Age timer

Forward Delay timers

Blocking

Forwarding

Discarding

Per-VLAN Spanning Tree (PVST)

PVST+

Rapid PVST+

Multiple Spanning Tree Protocol (MSTP)

Multiple Spanning Tree (MST)

PortFast

BPDU Guard